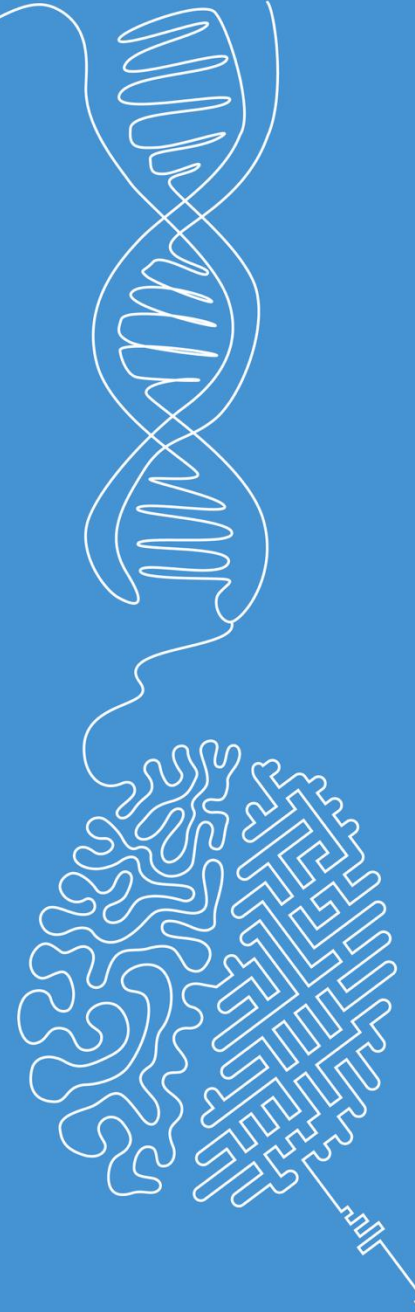


Fourier Series – Properties

Image and Signal Processing

Norman Juchler



Fourier series

Properties

Fourier series – Recap

Input:

- Real valued, periodic signals $x(t)$ with period T

Fourier series:

$$a_0 = \frac{1}{T} \int_{-T/2}^{+T/2} x(t) dt$$

$$a_k = \frac{2}{T} \int_{-T/2}^{+T/2} x(t) \cdot \cos\left(2\pi \frac{k}{T} t\right) dt, \quad \text{for } k \geq 1$$

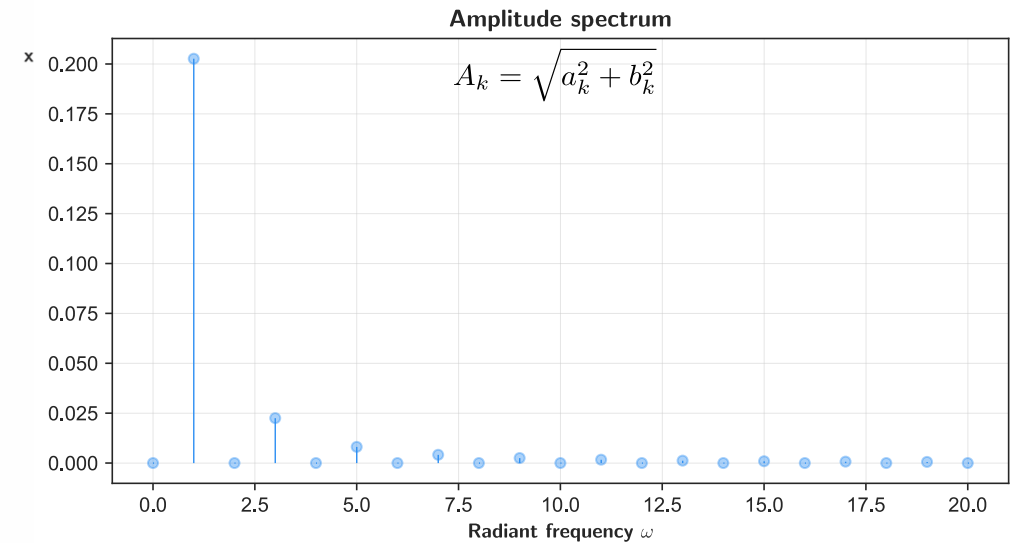
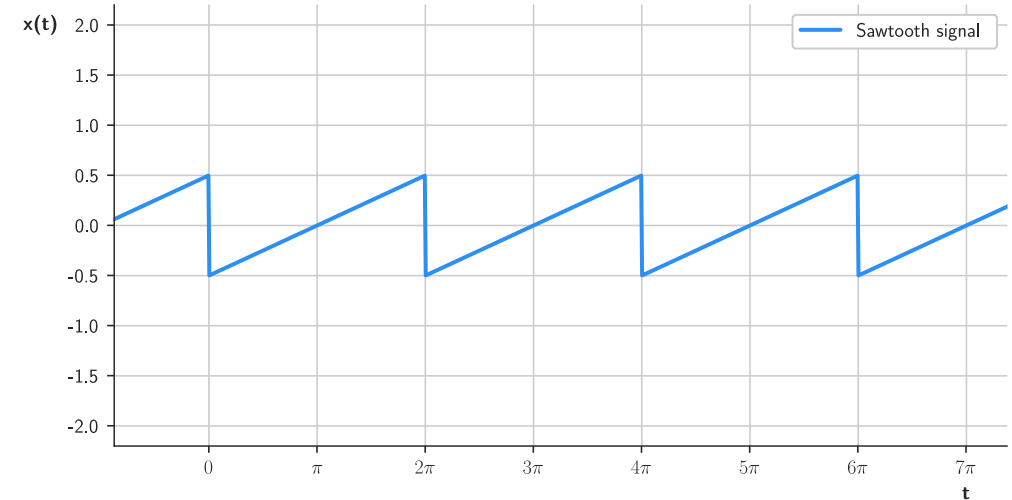
$$b_k = \frac{2}{T} \int_{-T/2}^{+T/2} x(t) \cdot \sin\left(2\pi \frac{k}{T} t\right) dt, \quad \text{for } k \geq 1$$

$$x(t) \approx x_N(t) = a_0 + \sum_{k=1}^N a_k \cos\left(\frac{2\pi k}{T} t\right) + b_k \sin\left(\frac{2\pi k}{T} t\right)$$

$\omega_k = 2\pi f_k = 2\pi \frac{k}{T}$

Output:

- Coefficients (e.g., a_k , b_k , and freqs. $\omega_k = 2\pi k/T$)
- Approximation $x_N(t)$



Fourier series – Recap

■ Input:

- Real valued, periodic signals $x(t)$ with period T

■ Fourier series:

$$\begin{aligned}
 a_0 &= \frac{1}{T} \int_{-T/2}^{+T/2} x(t) dt \\
 a_k &= \frac{2}{T} \int_{-T/2}^{+T/2} x(t) \cdot \cos\left(2\pi \frac{k}{T} t\right) dt, \quad \text{for } k \geq 1 \\
 b_k &= \frac{2}{T} \int_{-T/2}^{+T/2} x(t) \cdot \sin\left(2\pi \frac{k}{T} t\right) dt, \quad \text{for } k \geq 1 \\
 x(t) &\approx x_N(t) = a_0 + \sum_{k=1}^N a_k \cos\left(\frac{2\pi k}{T} t\right) + b_k \sin\left(\frac{2\pi k}{T} t\right)
 \end{aligned}$$

$\omega_k = 2\pi f_k = 2\pi \frac{k}{T}$

■ Output:

- Coefficients (e.g., a_k, b_k , and freqs. $\omega_k = 2\pi k/T$)
- Approximation $x_N(t)$

■ Useful results:

- Amplitude and phase spectra
- Symmetries simplify equations:
 - Odd functions $\rightarrow a_k = 0$
 - Even functions $\rightarrow b_k = 0$
- Superposition principle applies
- Energy is preserved between time and frequency domains:

~ Energy in the time domain

$$\int_{-T/2}^{T/2} |x(t)|^2 dt = 2\pi \sum_{k=-\infty}^{\infty} |c_k|^2 = \frac{1}{2} a_0^2 + \sum_{k=1}^{\infty} (a_k^2 + b_k^2)$$

(Parseval's identity)

~ Energy in the frequency domain

- Low frequencies describe global signal behavior; high frequencies encode fine details

Example: Rectangular pulse

Time-domain:

$$x_1(t) = \begin{cases} A, & |t| \leq \frac{r \cdot T}{2} \\ 0, & |t| > \frac{r \cdot T}{2} \end{cases}$$

This defines a rectangular pulse with period T , amplitude A , and a duty cycle $r \in [0, 1]$, which specifies the fraction of the period during which the signal is in the high state.

Frequency-domain:

$$a_0 = Ar$$

$$a_k = 2Ar \frac{\sin(\pi n r)}{\pi n r}$$

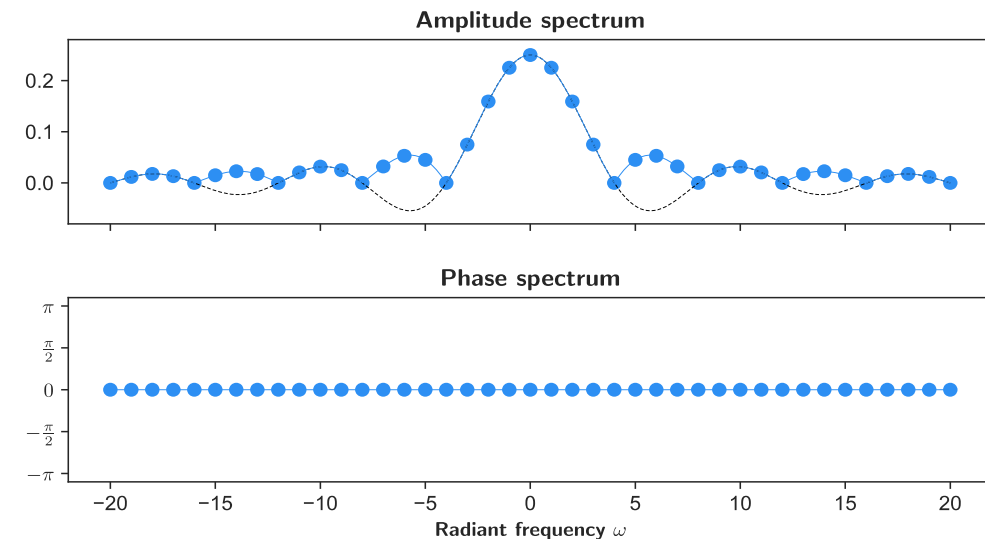
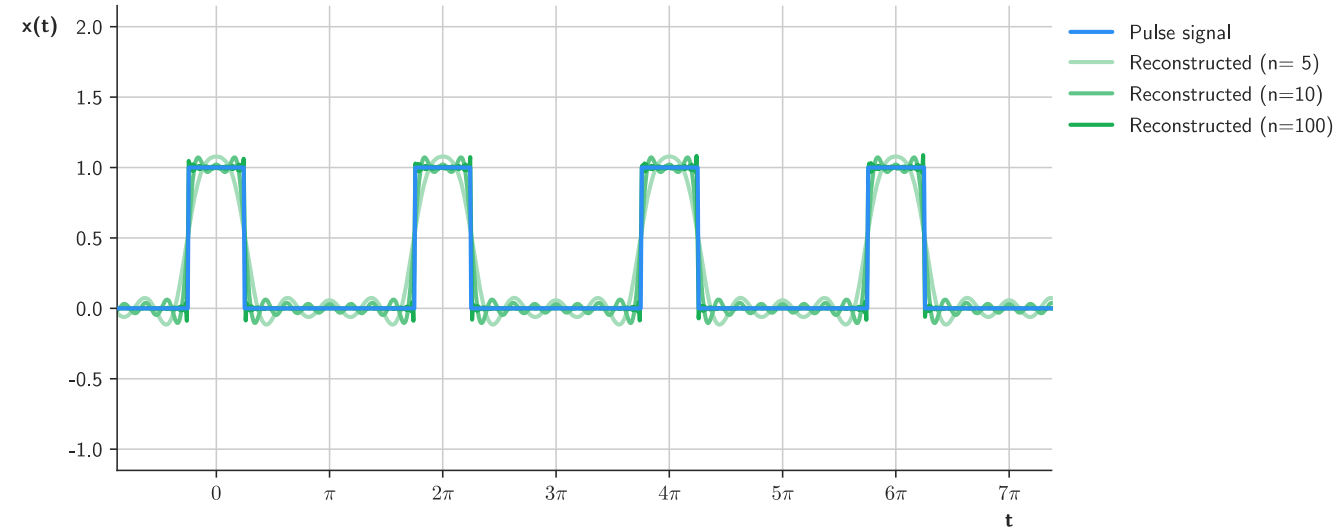
$$b_k = 0$$

$$\omega_k = \frac{2\pi k}{T}, \quad k > 0$$

Question:

How to compute the amplitude and phase spectrum from these coefficients?

Fourier coefficients of the periodic rectangular pulse as a function of its parameters. For their computation, see last week's Jupyter notebook. Note k only takes integral values.



Answer: Compare with last week's equivalence formulas.

$$A_k = \sqrt{a_k^2 + b_k^2}, \quad \varphi_k = \arctan \frac{b_k}{a_k}$$

Example: Rectangular pulse

Time-domain:

$$x_2(t) = \begin{cases} A, & 0 \leq t < r \cdot T \\ 0, & r \cdot T \leq t < T \end{cases}$$

A time-shifted version of $x_1(t)$:

$$x_2(t) = x_1\left(t - \frac{r}{2}\right)$$

Frequency-domain:

$$a_0 = Ar$$

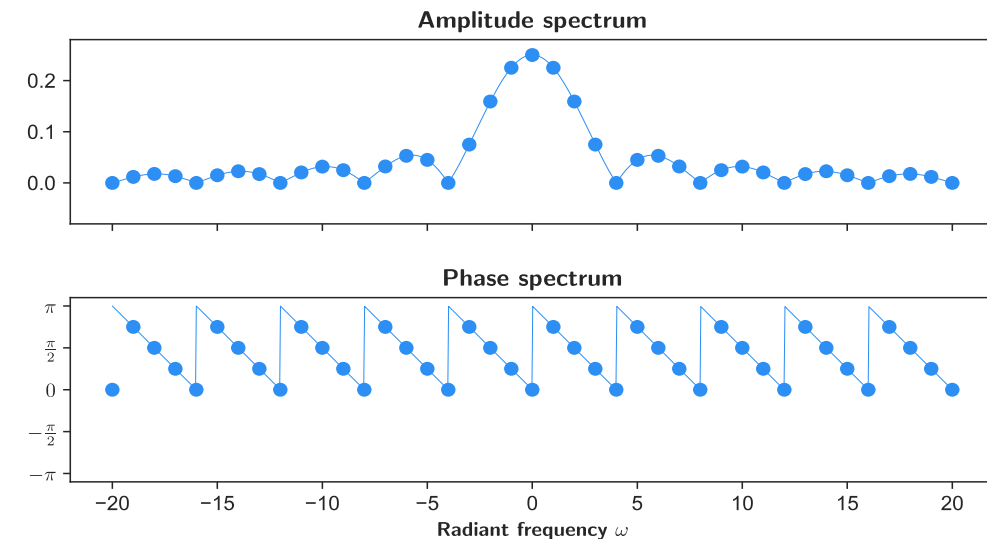
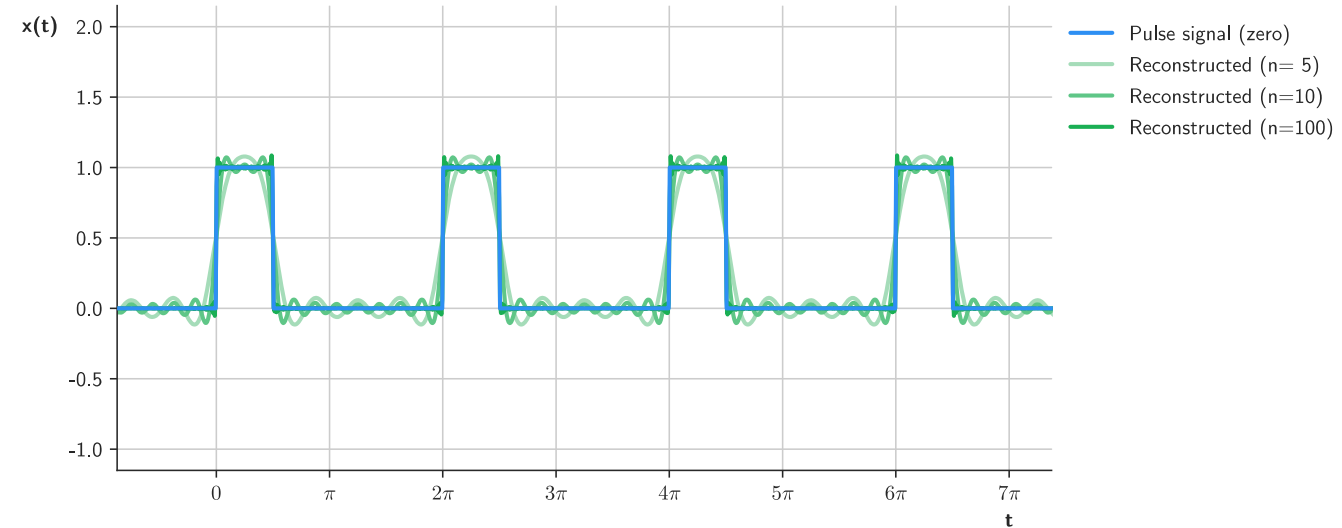
$$a_k = 2Ar \frac{\sin(2\pi kr)}{2\pi kr}$$

$$b_k = 2Ar \frac{\sin^2(\pi kr)}{\pi kr}$$

$$\omega_k = \frac{2\pi k}{T}, \quad k > 0$$

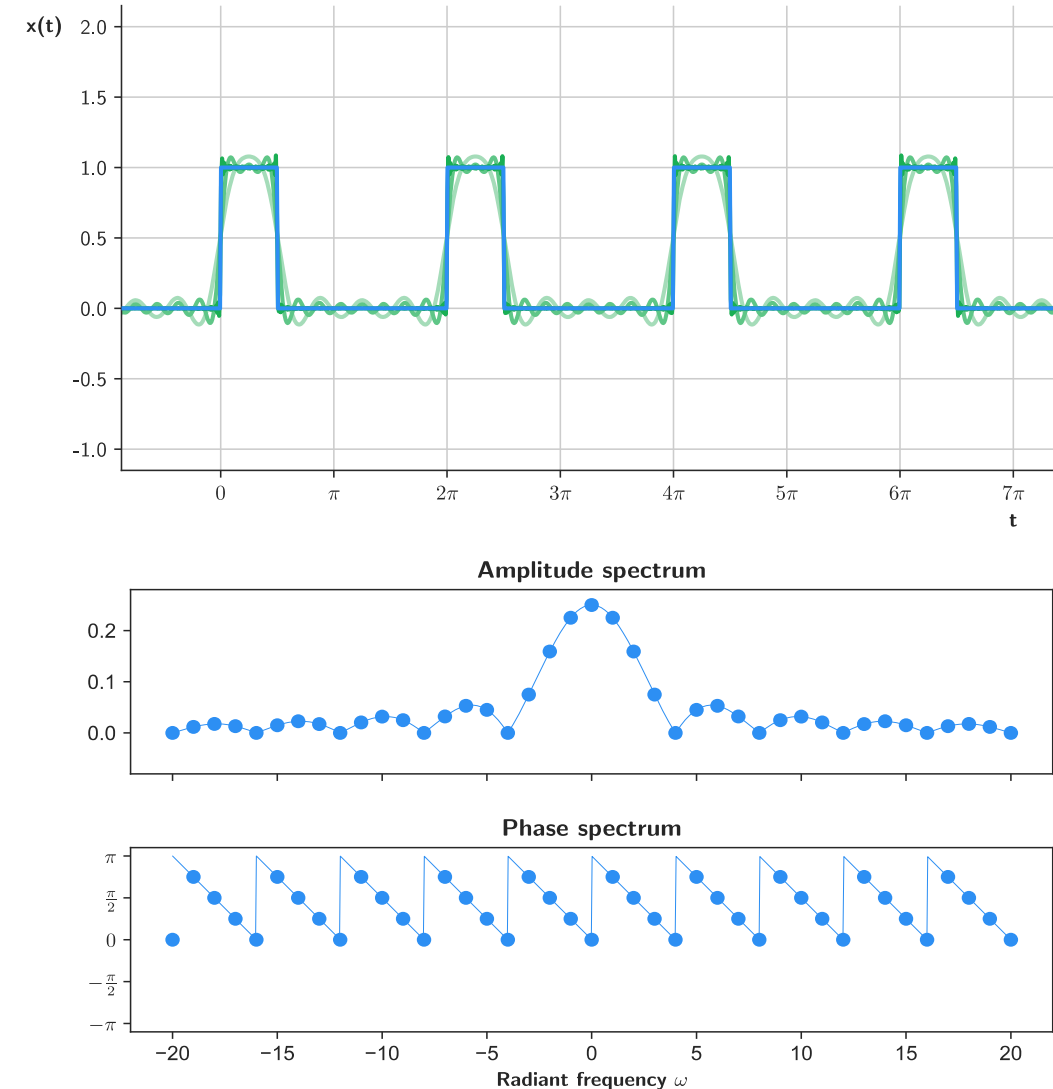
Compare the Fourier coefficients, and the amplitude and phase spectra of $x_1(t)$ and $x_2(t)$:

Although both a_k and b_k change, the time shift affects only the phase, not the amplitude. This time-shift property (shift $\rightarrow$ phase) holds for all Fourier transforms we will encounter.



Key insights

- We can express arbitrary periodic signals as weighted sums of sines and cosines.
- The Fourier coefficients of an input signal are obtained by evaluating definite integrals (see Jupyter notebook for examples).
- Symmetry plays a role!
 - a_k are zero for odd functions
 - b_k are zero for even functions
- Fourier series produce **discrete** frequency-domain representations (i.e., both the amplitude and phase spectra are discrete).



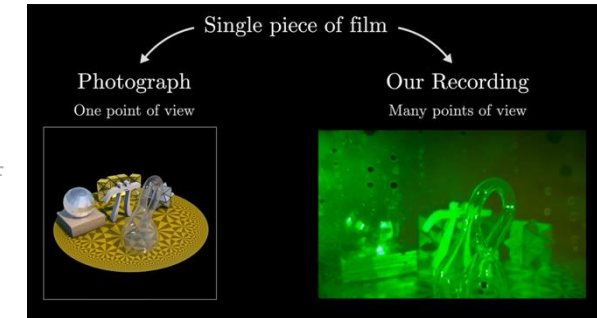
Key insights

- A time shift affects only the phase, not the amplitude.
- The phase of a signal encodes timing information (shift) and localization of features (peaks, edges of a pulse, corners) in the time domain.
- The **sinc function** appears naturally in the Fourier coefficients and is fundamental enough to merit its own name.

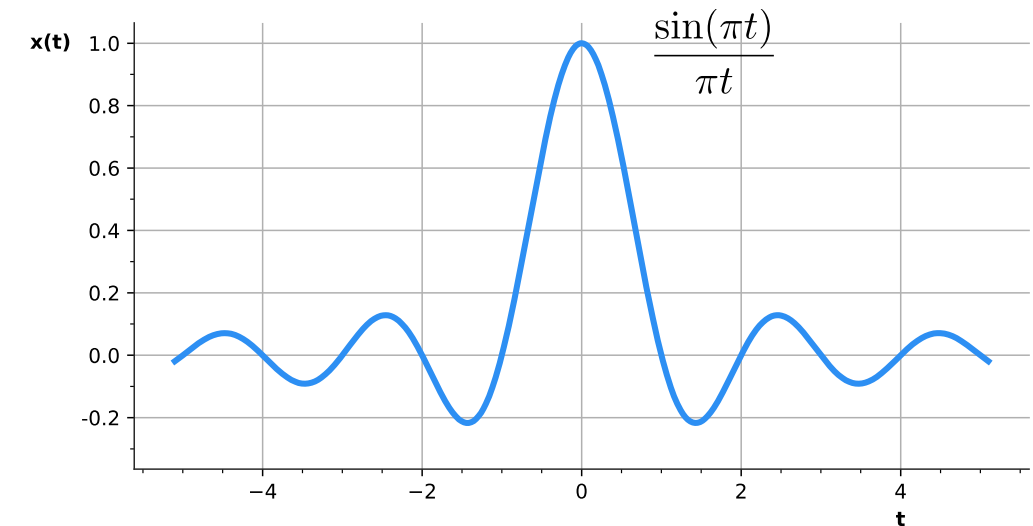
$$x(t) = \text{sinc}(t) = \frac{\sin(\pi t)}{\pi t}$$
$$x(0) = \text{sinc}(0) = 1$$

Excursion:

Holograms provide an impressive and illustrative example of how crucial phase information is in certain applications.



Video: [YouTube / 3blue1brown](https://www.youtube.com/watch?v=3blue1brown)



Key insights

Limitation: Fourier series apply only to periodic signals!

Question: Can we generalize this to aperiodic signals?

